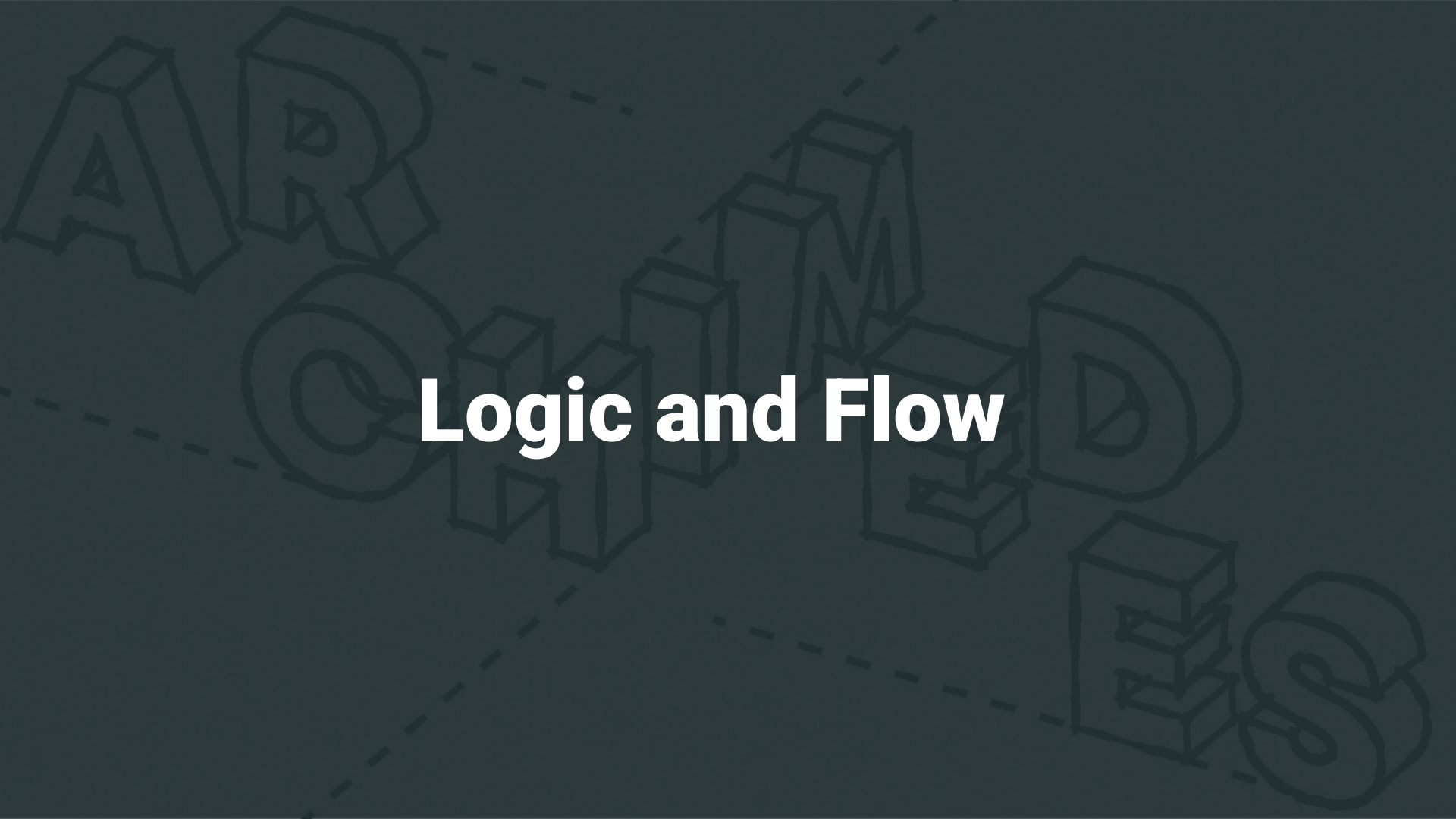




The Auto-Scheduler Problem

June 2, 2025

LUMA Development Slides



Logic and Flow

Inputs

1. A list of all days and time range for interviews
2. A list of all *applicants* and their availability

John Doe	Sept. 10	Sept. 11	Sept. 12
	1-2 PM, 2-3 PM, 5-6 PM	10-11 AM, 2-3 PM	6-7 PM, 7-8 PM

3. A list of all *recruiters* and their availability

Grant Chapman	Sept. 10	Sept. 11	Sept. 12
		10-11 AM	1-2 PM, 2-3 PM, 3-4 PM

4. A list of all recruiters and their preferred team(s)
5. A list of all *individual* interview rooms and the times they are reserved for
6. A list of all *group* interview rooms and the times they are reserved for
7. A list of all group interview prompt names

Outputs

1. For each applicant:
 - a. Date and time of interview
 - b. Room number
 - c. Group letter
 - d. Assigned interviewer
2. Any applicants that were not scheduled
3. For each recruiter:
 - a. Their cohort
 - b. The cohort they are standing-by for
 - c. Their schedule for each day
 - i. Dates and times
 - ii. Room numbers
 - iii. Group letters
 - iv. Assigned interviewees
4. For each group, an assigned prompt

Constraints

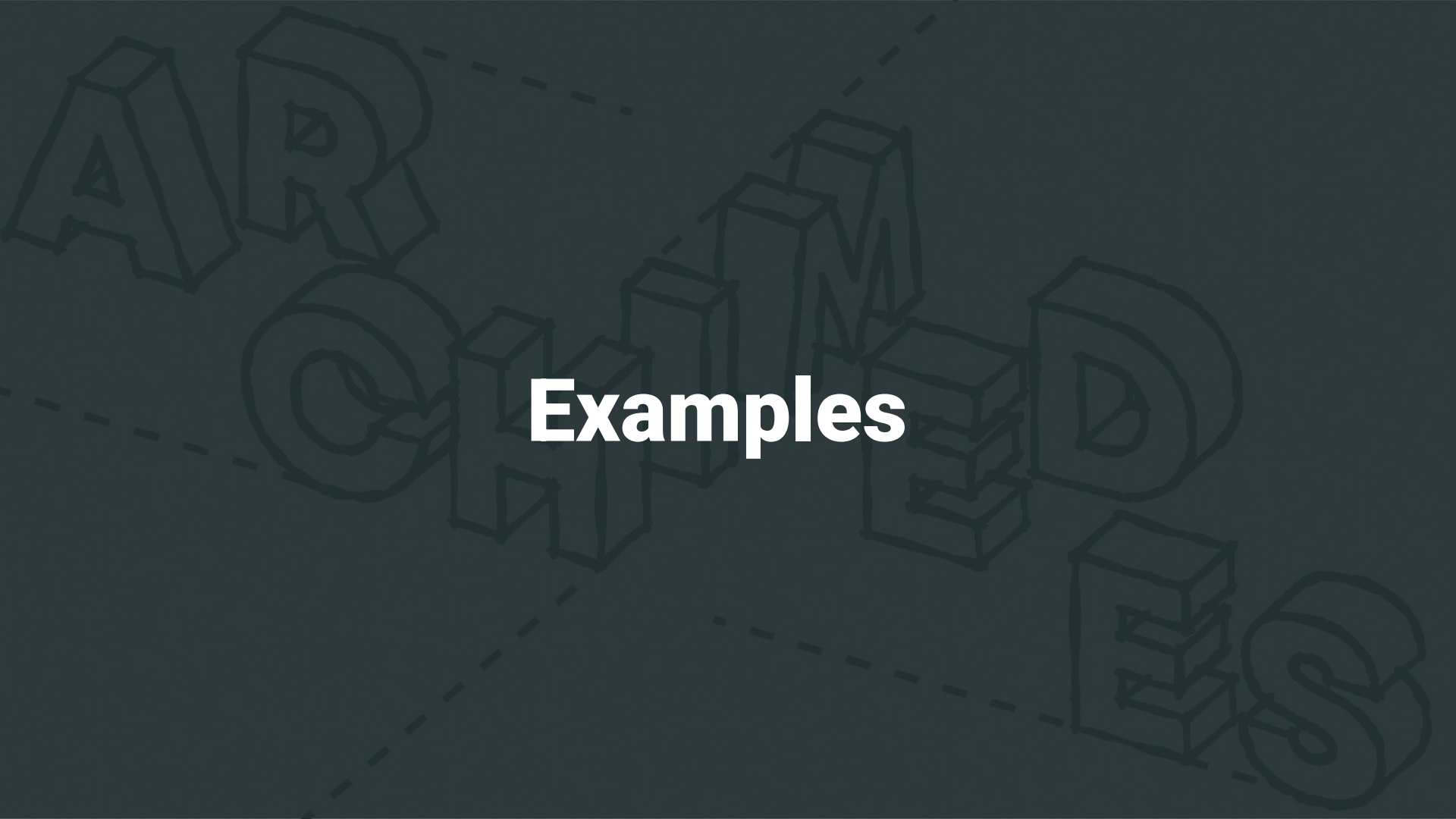
- Rooms can only be occupied one at a time
- A recruiter can only be in one room at a time
- Minimum and maximum group interview size
 - According to Kevin's SOP, target 8 (or 4, at your discretion) interviewees
- Maximum simultaneous interviews
 - May be variable depending on how many recruiters sign-up for interviews
- Maximum break gap for a given recruiter
 - For example, a 2-hour gap between interviews is not desirable
- Individual interviews are 20 minutes long and group interviews are 30 minutes long
 - These times include transition
- Applicants should have their individual and group interviews back-to-back (or at least very close to this)

Alterables

- Availability selection increment
 - For example, select availability by increments of 2 hours instead of 1 hour
- Layout and timing of blocks and slots
- Cohort logistics (consult Kevin about changes). In its current version:
 - 8 applicants per cohort (may want to consider 4)
 - Recruiters are a part of only *one* cohort and standby for only *one other* cohort

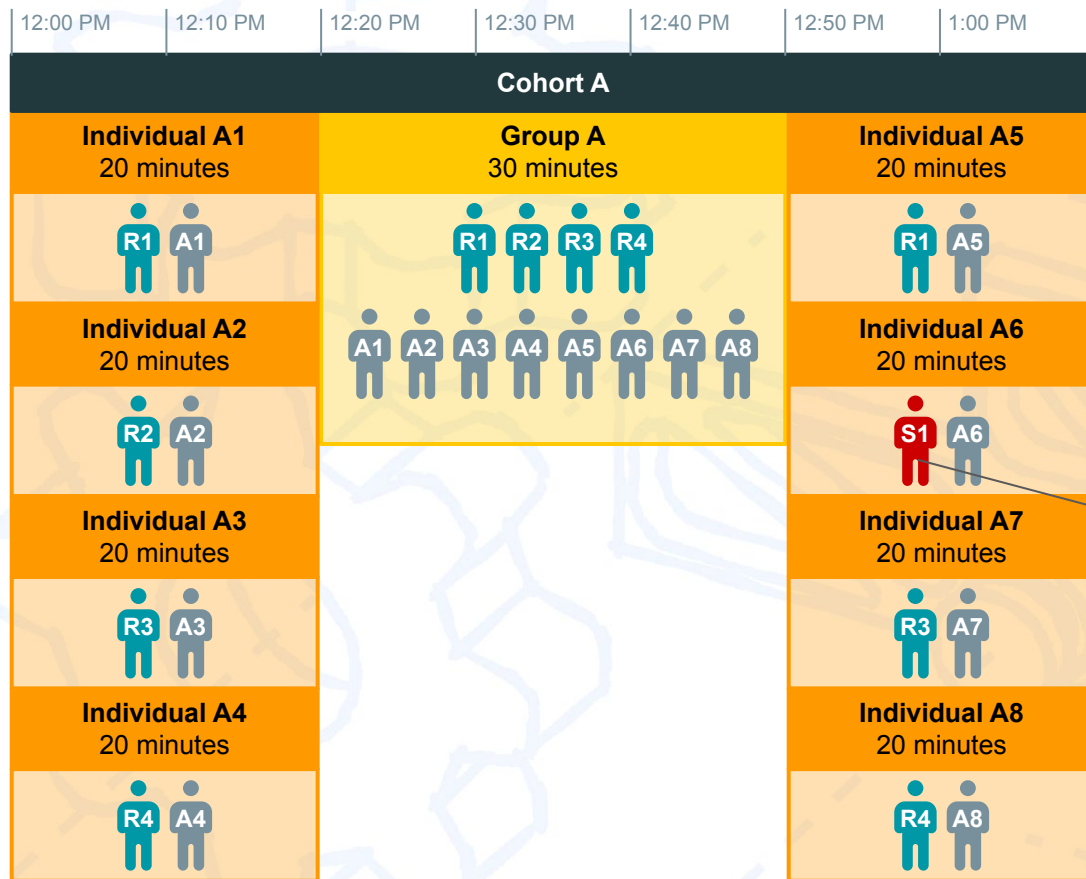
Preferences

- Cohorts should follow Kevin's SOP
- Recruiters should be paired with their preferred team(s)
 - If no recruiter can be paired with a given applicant, this may be loosened
- Ability to add time exclusions
 - For example, 12 – 1 PM may be excluded for all recruiters as lunch time

The background features a dark gray surface with several 3D, blocky letters in a lighter gray color. The letters are arranged in a way that suggests they are floating or scattered. Dashed white lines are drawn across the background, creating a sense of perspective and depth. The word "Examples" is centered in the middle of the image in a bold, white, sans-serif font.

Examples

Example Timeline (Zoomed In)



Here, R2 was unavailable, so S1 from another cohort (who was standing-by for Cohort A) filled in

Example Timeline (Zoomed Out)

Variation 1: Sequential Timeline

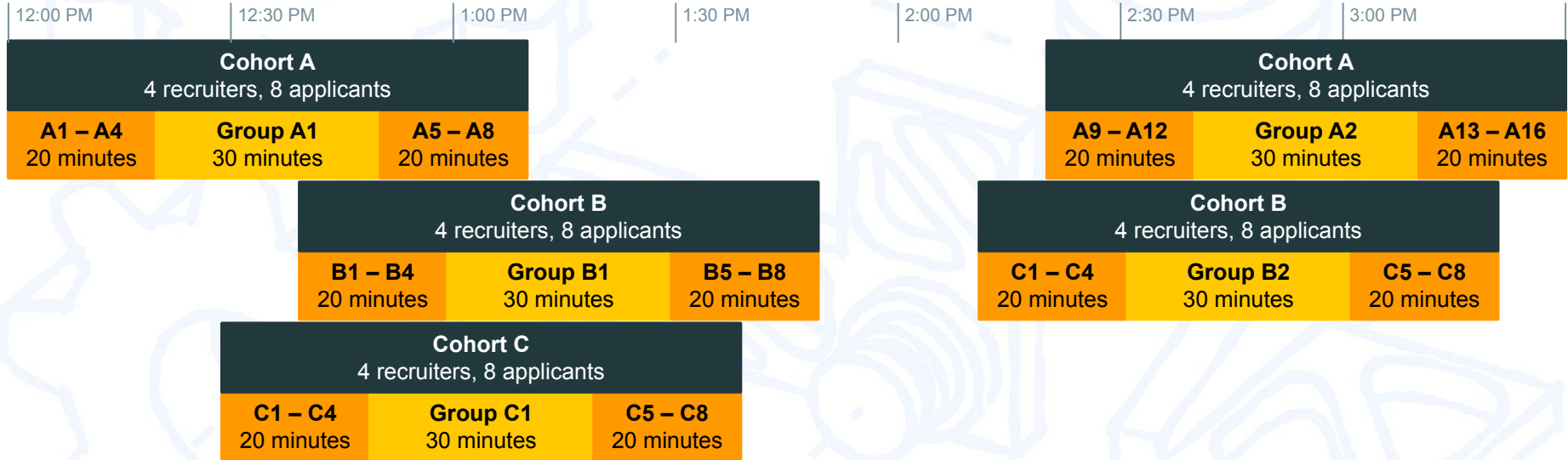
Cohorts are limited to defined blocks.

12:00 PM	12:30 PM	1:00 PM	1:30 PM	2:00 PM	2:30 PM	3:00 PM
Cohort A 4 recruiters, 8 applicants			Cohort A 4 recruiters, 8 applicants			Cohort A 4 recruiters, 8 applicants
A1 – A4 20 minutes	Group A1 30 minutes	A5 – A8 20 minutes	A9 – A12 20 minutes	Group A2 30 minutes	A13 – A16 20 minutes	A17 – A20 20 minutes
Cohort B 4 recruiters, 8 applicants			Cohort B 4 recruiters, 8 applicants			
B1 – B4 20 minutes	Group B1 30 minutes	B5 – B8 20 minutes	B9 – B12 20 minutes	Group B2 30 minutes	B13 – B16 20 minutes	
			Cohort C 4 recruiters, 8 applicants			Cohort C 4 recruiters, 8 applicants
			C1 – C4 20 minutes	Group C1 30 minutes	C5 – C8 20 minutes	C9 – C12 20 minutes
						Group C2 30 minutes
						C13 – C16 20 minutes

Example Timeline (Zoomed Out)

Variation 2: Variable Timeline

Cohorts are placed only where they are needed and available.



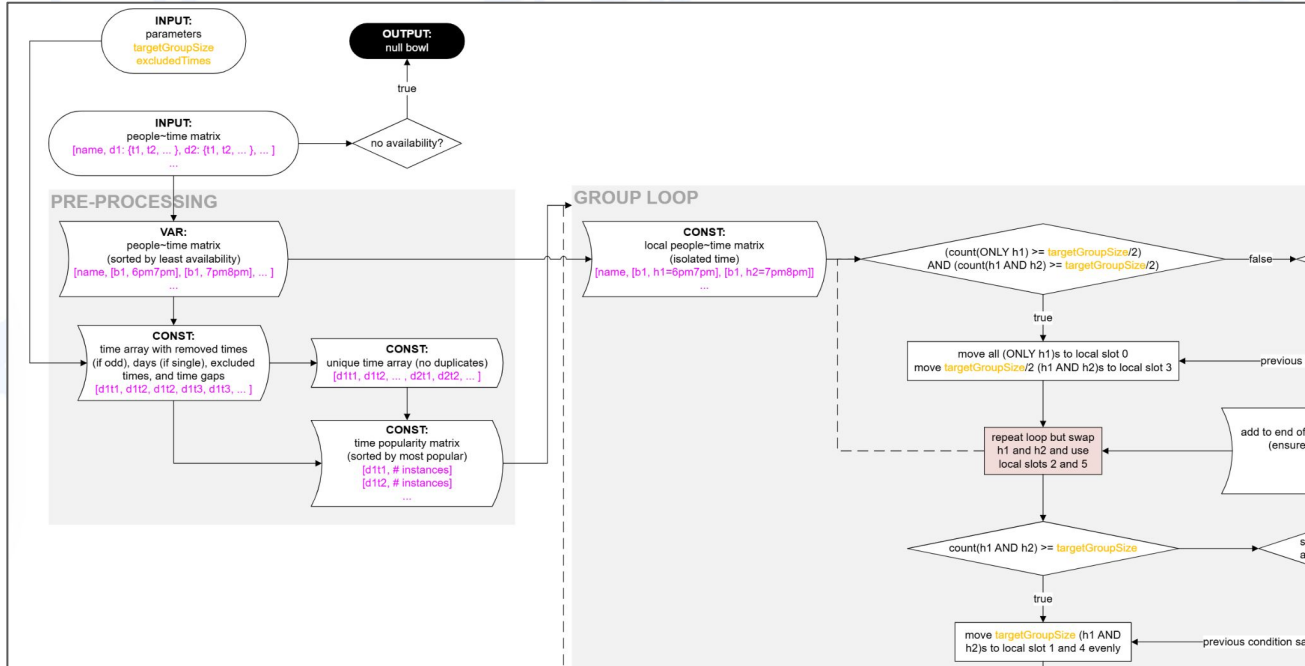
The background features the word "ARCHITECTS" in large, 3D, blocky letters. A dashed line runs diagonally across the image from the top left to the bottom right. The word "Tasks" is centered in the middle of the image.

Tasks

Logic and Flow

Develop a flow chart which outlines how everything is scheduled

- *Need this before attempting any code!*



Microsoft Visio example

Programming

- Back-end
 - Code the auto-scheduler and test with sample data
 - Consider issues such as capacity and recruiter shortages
- Front-end
 - Input area for interview days and time ranges
 - Input area for individual and group room reservations and the times they are available
 - Input area for parameters such as minimum/maximum group size, time exclusions, and group interview prompt names
 - Input area for recruiter availability and preferred teams
 - Final schedule layout